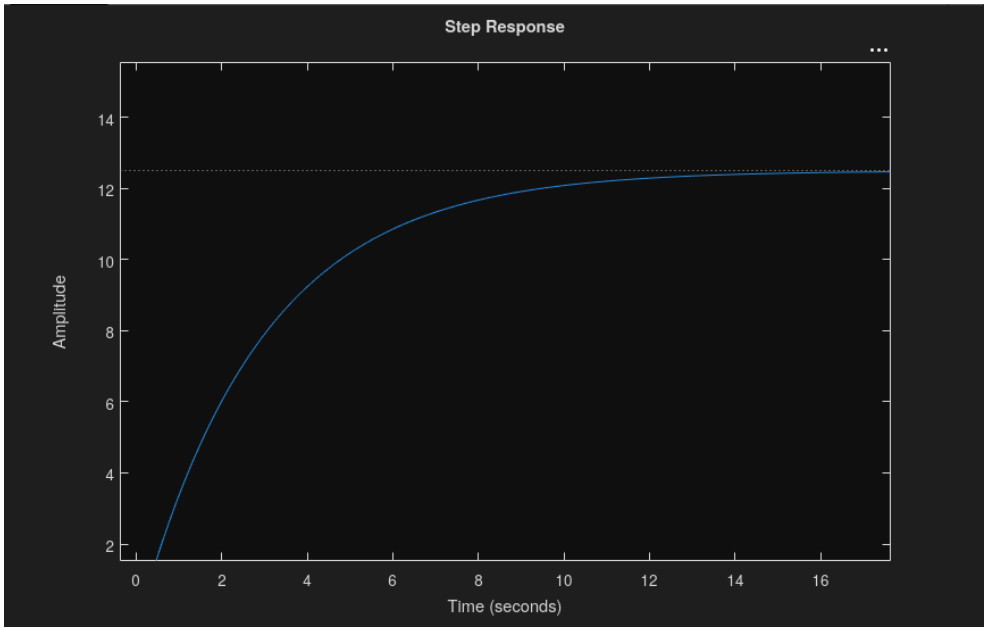
11)

A/B:



C:

Peak value: 12.5
Peak time: 19.3 seconds
Rise time: time at 90% (11.25) - 10%(1.25) = 6.65 - 0.397 = 6.253
Settling time: 12.4

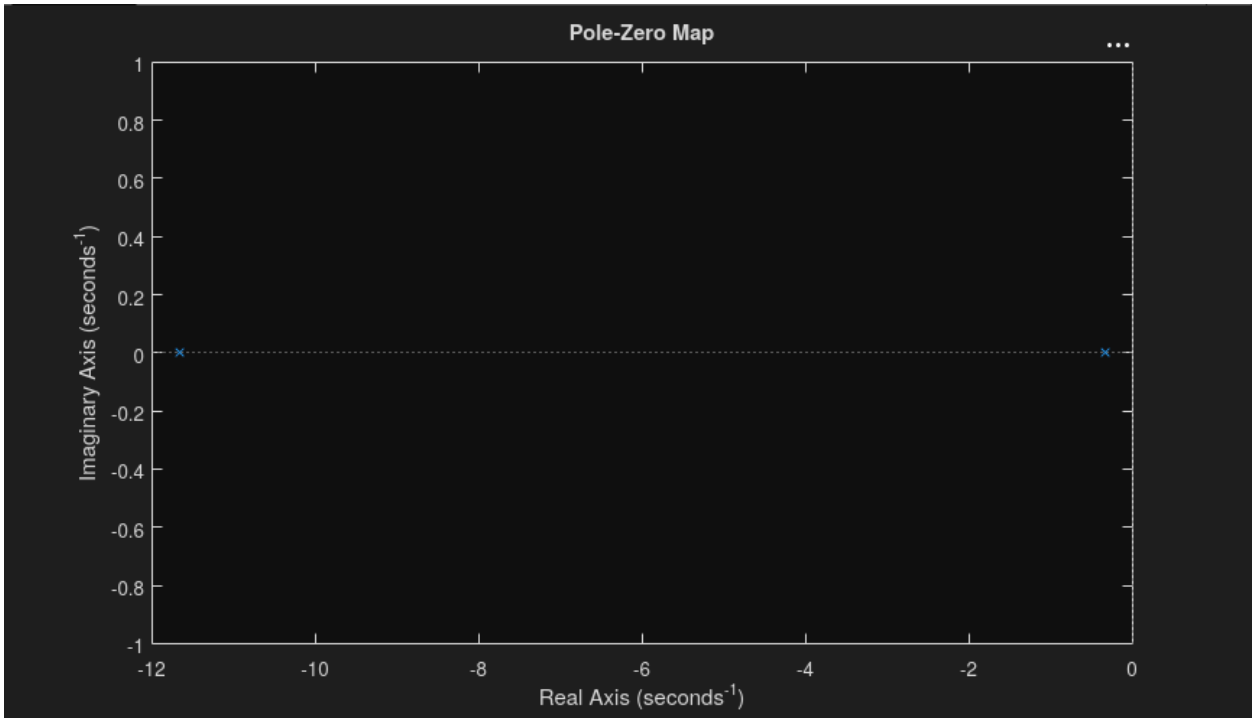
D:

Peak value: 12.4915
Peak time: 21.335
Rise time: 6.4048
Settling time: 11.488

E:

I was fairly close with the peak value and rise time, however peak time and settling time were off by a decent amount. It's difficult to eyeball the time values.

F:

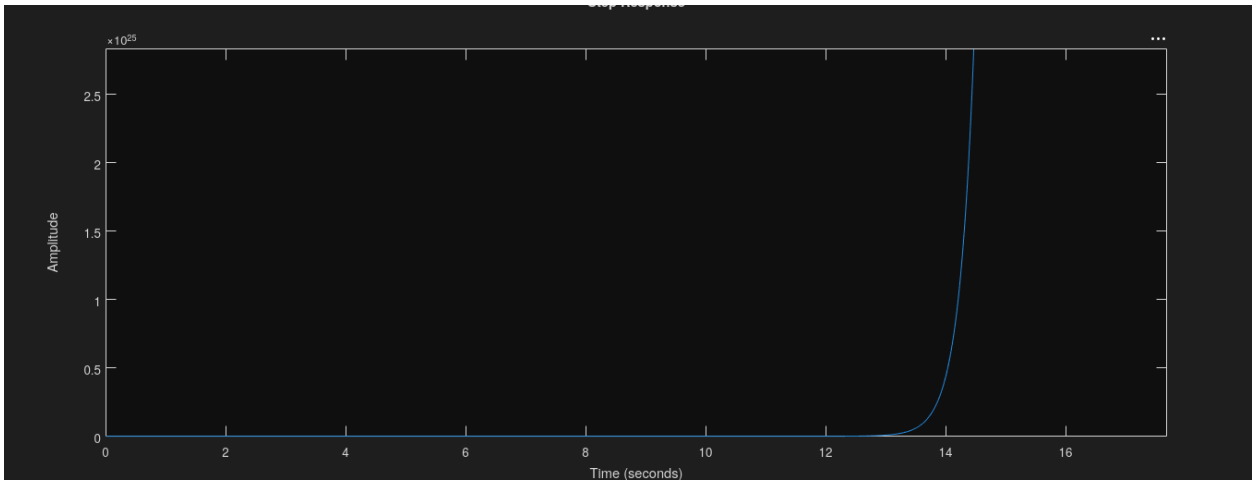


G:

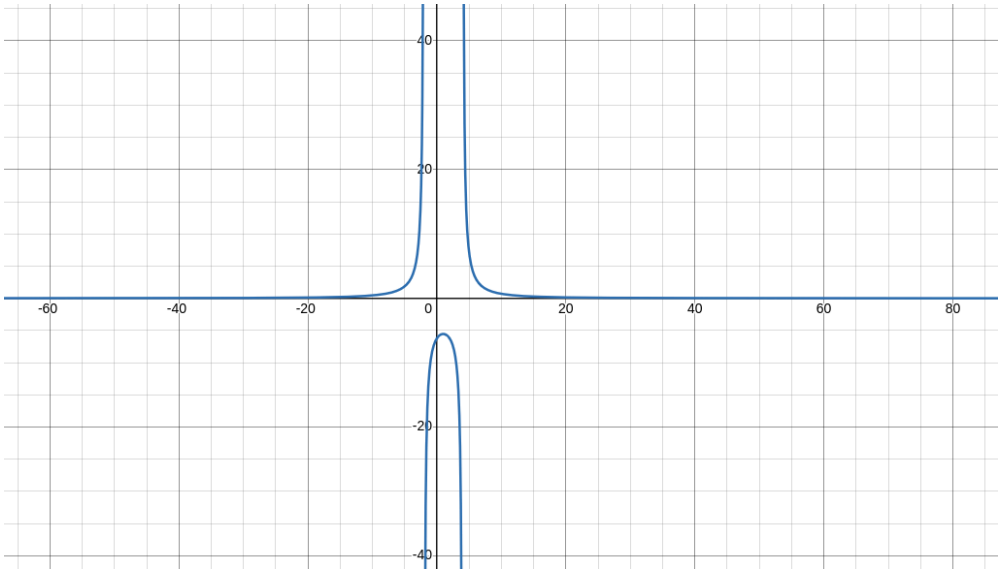
The system is stable since there are no RHPs in the pole zero map. The step response graph near perfectly reaches the its ideal point without cutoff or oscillation so it is critically damped

12)

A/B:



No matter how I messed with it I could not get it to reach a steady state. When I put the same graph in desmos this is what i get:



C:

Peak value is infinite
Peak time is unavailable
Rise time and settling time is unavailable

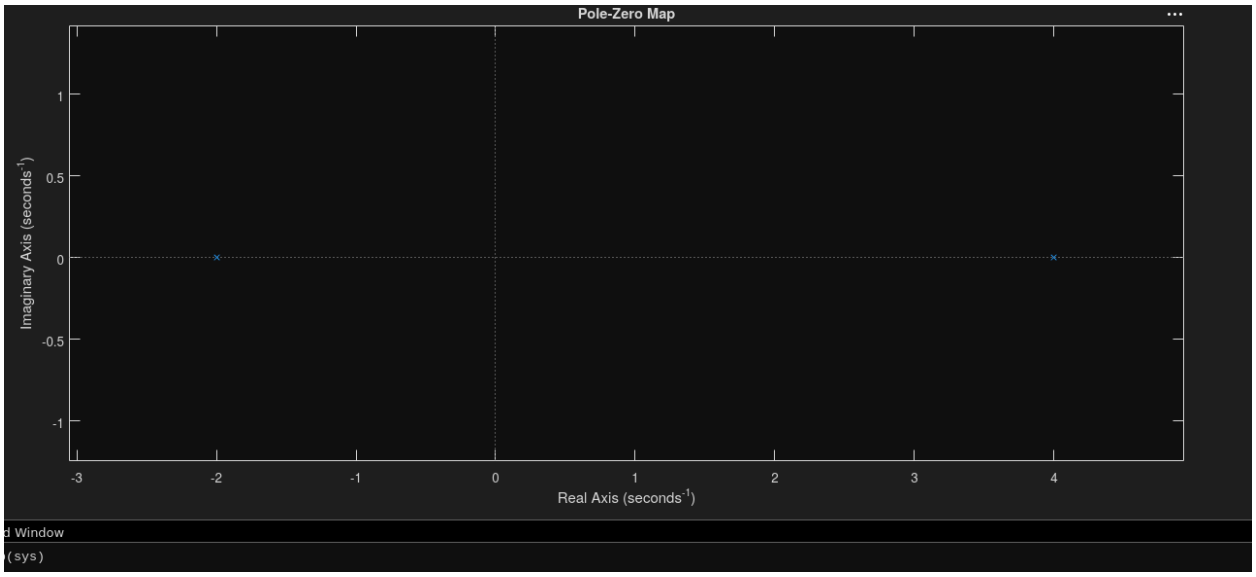
D:

Peak value: infinite
Peak time: infinite
Rise time: nan
Settling time: nan

E:

There is nothing to really compare. The results are either infinite or unavailable

F:

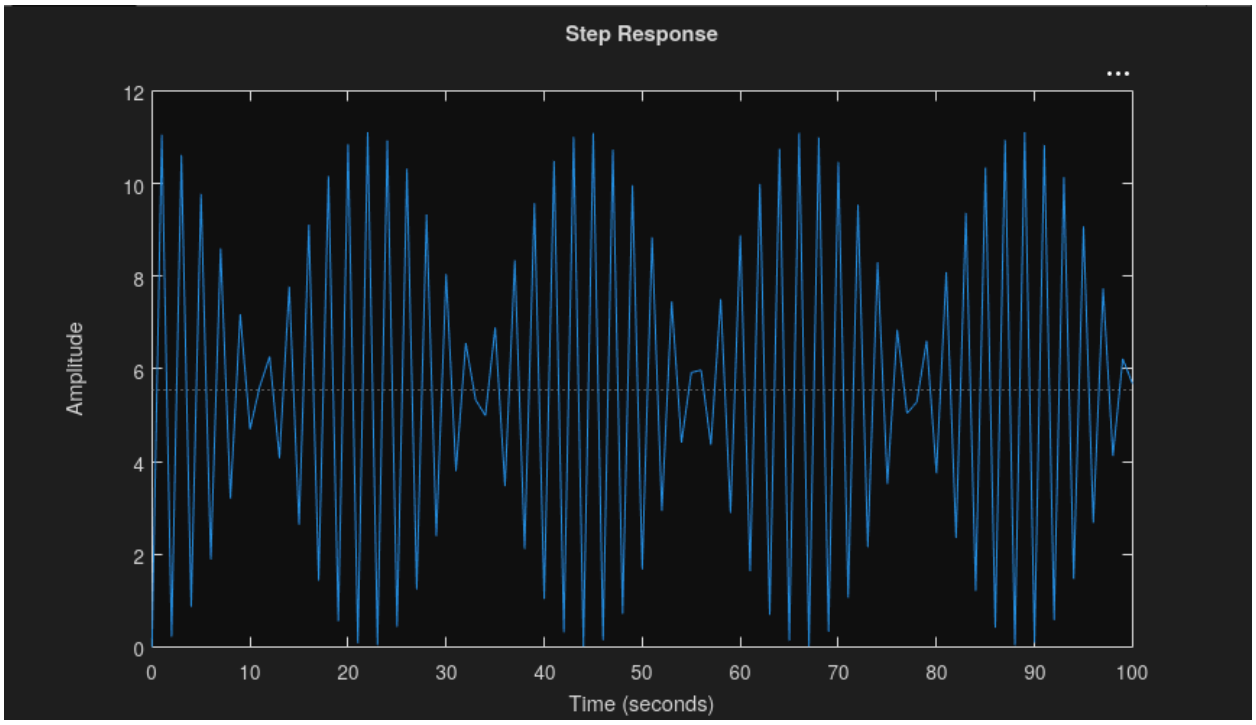


G:

The graph is unstable because there is a RHP in the pole zero map. This instability leads to there being nothing available for its dempening.

13)

A/B:



C:

No values available

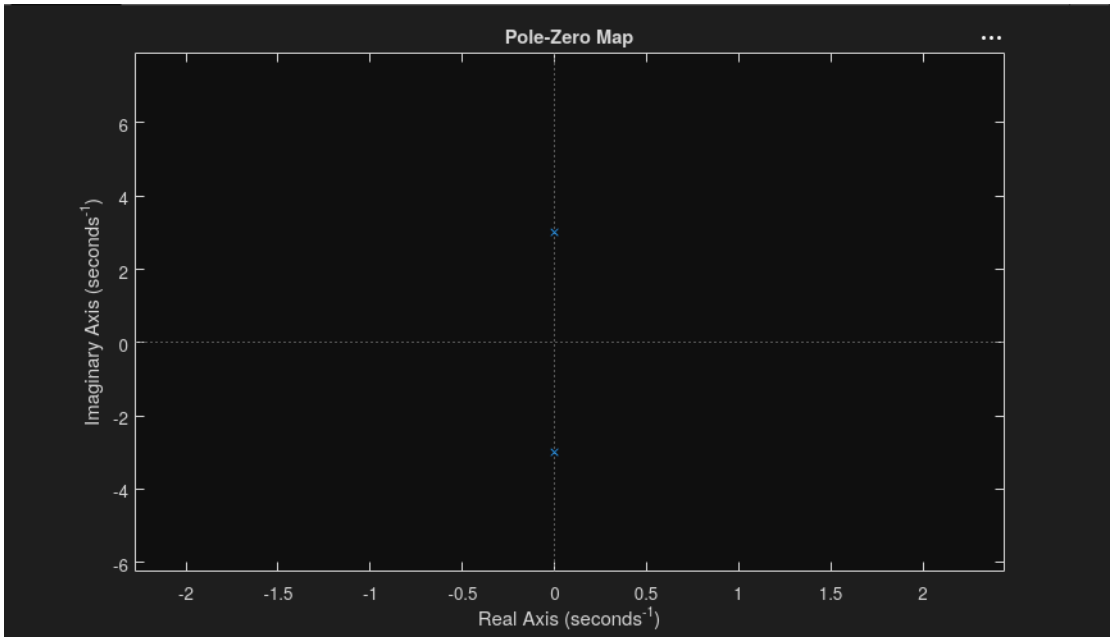
D:

Peak value: Nan
Peak time:nan
Rise time: nan
Settling time: nan

E:

No proper results that can be compared

F:



G:

This system has marginal stability and is undamped. The poles are on the y-axis causing the marginal stability, and the graph itself never reaches a proper point, implying no damping at all.